

100 Days of Linux - Day 099

Title: Bash Scripting Mini Project

Today's Goal

Combine everything you've learned so far to build a practical Bash scripting project.

Why This Matters

Real-world Bash scripts combine variables, user input, conditions and loops to automate repetitive tasks.

Project: Student Information Manager

Requirements

Your script should:

1. Ask for the student's name using **read**.
2. Ask for the student's age.
3. Ask how many subjects to enter.
4. Use a **for** loop to read each subject name.
5. Use an **if** statement to check whether the student is 18 or older.
6. Display a formatted summary of all entered information.

Suggested Script Structure

- Shebang (`#!/bin/bash`)
- Variables
- `read` commands
- `for` loop
- `if / else` statement
- Final summary using `echo`

Practice Tasks

1. Save the script as **student_manager.sh**.
2. Make it executable with **chmod +x**.
3. Run it multiple times with different inputs.
4. Improve the output formatting.
5. Add one extra feature of your choice.

Hints

Break the project into small steps and test each section before adding the next one.

Common Mistakes

Trying to write the entire script at once, forgetting to initialise variables or placing loop logic outside the loop.

Did You Know?

Many system administration scripts start as small utilities like this before growing into powerful automation tools.

Pro Tip

When a script becomes long, add comments to explain each section for future maintenance.

Instructor Checklist

- ✓ Uses variables
- ✓ Uses `read`
- ✓ Uses a `for` loop
- ✓ Uses an `if/else` statement
- ✓ Produces the expected output

Homework

Enhance the project by allowing the user to run it repeatedly until they choose to exit.

Next Day Preview

Tomorrow you'll complete the 100 Days of Linux journey with a final capstone challenge and course wrap-up.